



COMSATS University
Islamabad
Abbottabad Campus

Department of Computer Science

SUBJECT:

SOFTWARE QUALITY ENGINEERING

LAB TASK 4.

SUBMITTED TO:

SIR MUKHTIAR ZAMIN

SUBMITTED BY:

HAMZA ALI FA23-BSE-142

SECTION:

BSE – 6B

DATE:

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Lab Tasks 1.

Write a program in JAVA to solve the triangle problem defined as follows: Accept three integers which are supposed to be the three sides of a triangle and determine if the three values represent an equilateral triangle, isosceles triangle, scalene triangle, or they do not form a triangle at all. Write unit test cases for the triangle class. Which will check the functionality of triangle class

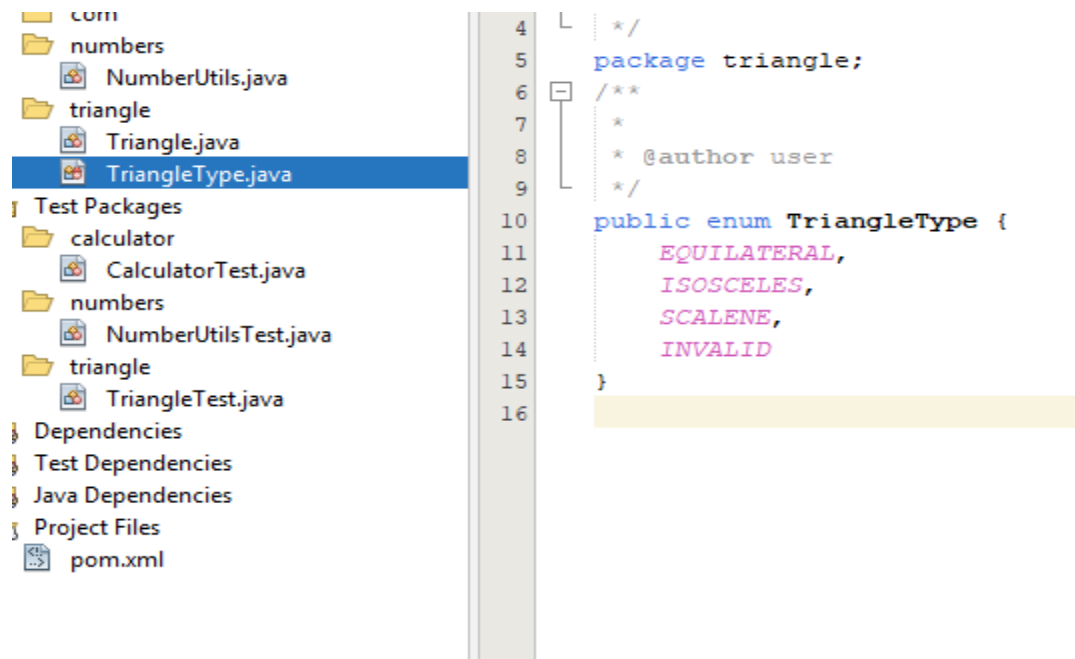
Solution:

Triangle java class

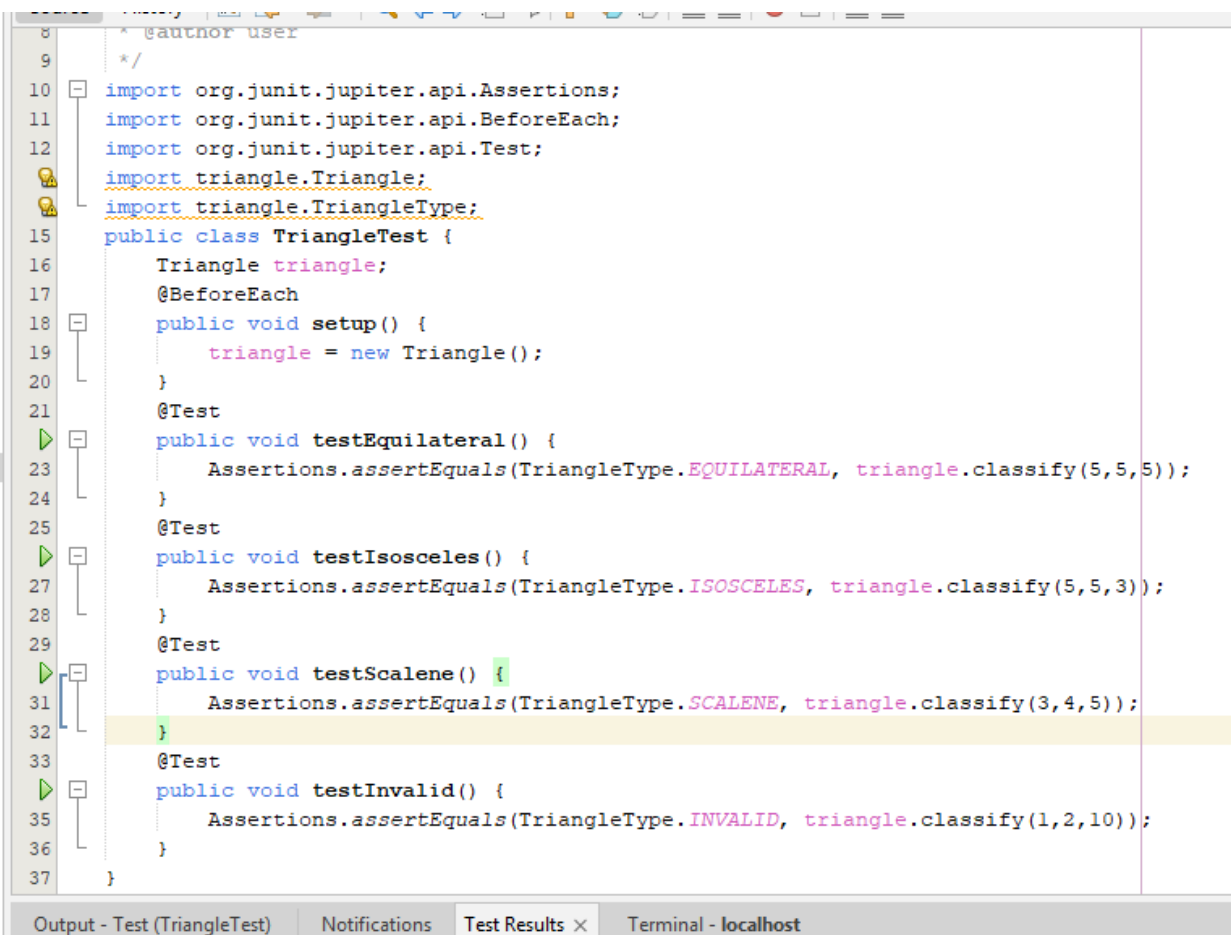


```
5 package triangle;
6 /**
7  *
8  * @author user
9  */
10 public class Triangle {
11     public TriangleType classify(int a, int b, int c) {
12         // Invalid condition
13         if (a <= 0 || b <= 0 || c <= 0) {
14             return TriangleType.INVALID;
15         }
16         if (a + b <= c || a + c <= b || b + c <= a) {
17             return TriangleType.INVALID;
18         }
19         // Equilateral
20         if (a == b && b == c) {
21             return TriangleType.EQUILATERAL;
22         }
23         // Isosceles
24         if (a == b || b == c || a == c) {
25             return TriangleType.ISOSCELES;
26         }
27         // Scalene
28         return TriangleType.SCALENE;
29     }
30 }
31 }
```

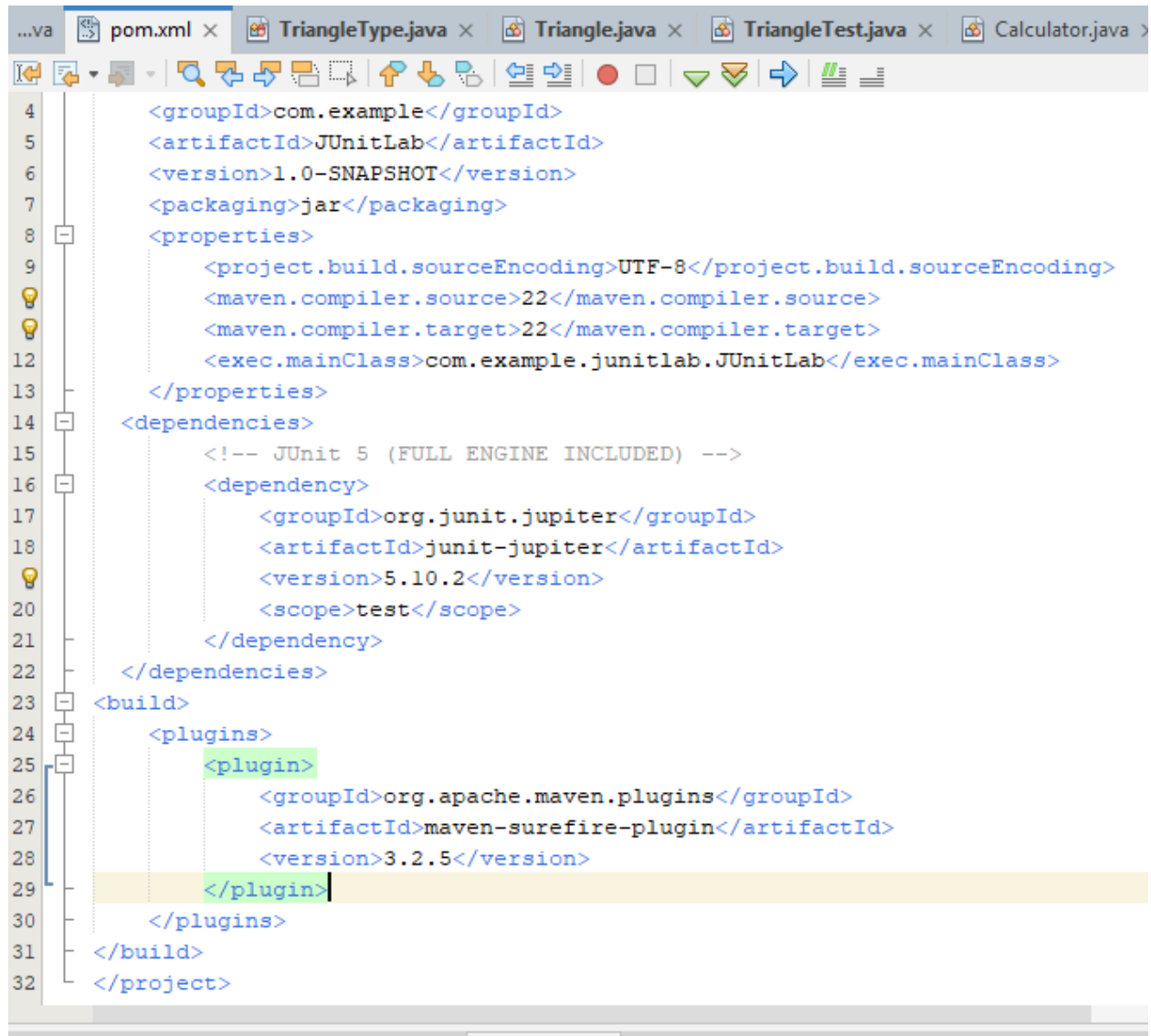
Triangle type java class



Triangle test java class

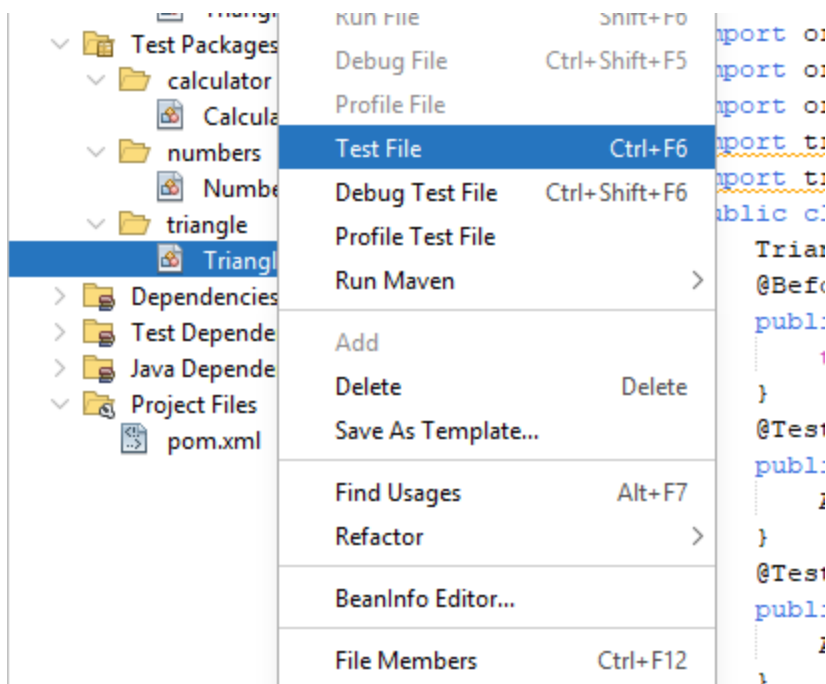


Pom.xml file code



```
4 <groupId>com.example</groupId>
5 <artifactId>JUnitLab</artifactId>
6 <version>1.0-SNAPSHOT</version>
7 <packaging>jar</packaging>
8 <properties>
9     <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
10    <maven.compiler.source>22</maven.compiler.source>
11    <maven.compiler.target>22</maven.compiler.target>
12    <exec.mainClass>com.example.junitlab.JUnitLab</exec.mainClass>
13 </properties>
14 <dependencies>
15     <!-- JUnit 5 (FULL ENGINE INCLUDED) -->
16     <dependency>
17         <groupId>org.junit.jupiter</groupId>
18         <artifactId>junit-jupiter</artifactId>
19         <version>5.10.2</version>
20         <scope>test</scope>
21     </dependency>
22 </dependencies>
23 <build>
24     <plugins>
25         <plugin>
26             <groupId>org.apache.maven.plugins</groupId>
27             <artifactId>maven-surefire-plugin</artifactId>
28             <version>3.2.5</version>
29         </plugin>
30     </plugins>
31 </build>
32 </project>
```

For testing triangle test



Output

```

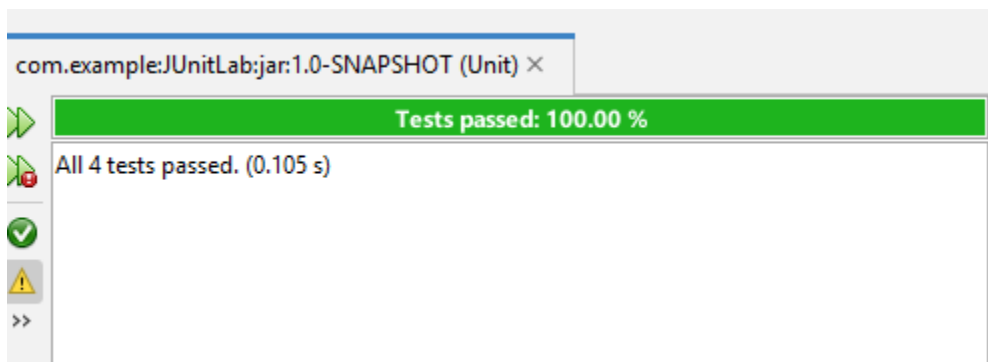
-----
Running triangle.TriangleTest
Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.105 s -- in triangle.TriangleTest

Results:

Tests run: 4, Failures: 0, Errors: 0, Skipped: 0

-----
BUILD SUCCESS
-----
Total time: 2.468 s
Finished at: 2026-04-25T22:10:03+05:00
-----

```



2. Write calculator class having methods addition, multiplication , subtract, Square, Cube.
Write testcases to test the functionality of calculator class.

Calculator java class

```
4  */
5  package calculator;
6  /**
7   *
8   * @author user
9   */
10 public class Calculator {
11     public int add(int a, int b) {
12         return a + b;
13     }
14     public int subtract(int a, int b) {
15         return a - b;
16     }
17     public int multiply(int a, int b) {
18         return a * b;
19     }
20     public int square(int a) {
21         return a * a;
22     }
23     public int cube(int a) {
24         return a * a * a;
25     }
26 }
27
```

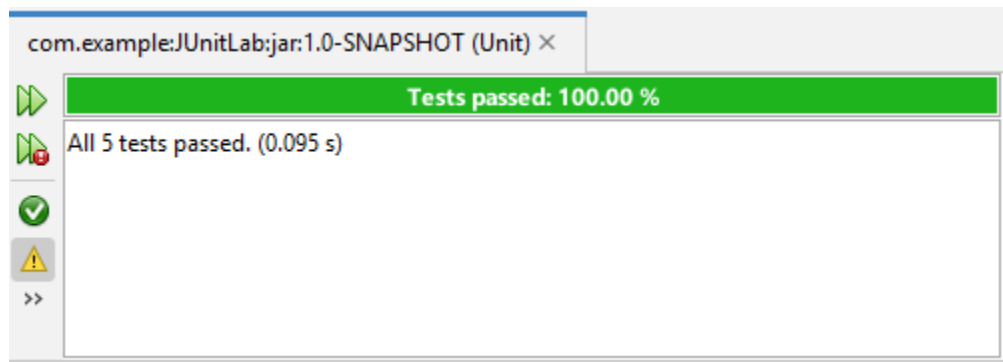
Calculator test java class

```
10 import org.junit.jupiter.api.Assertions;
11 import org.junit.jupiter.api.BeforeEach;
12 import org.junit.jupiter.api.Test;
13 public class CalculatorTest {
14     Calculator calc;
15     @BeforeEach
16     public void setup() {
17         calc = new Calculator();
18     }
19     @Test
20     public void testAddition() {
21         Assertions.assertEquals(15, calc.add(10,5));
22     }
23     @Test
24     public void testSubtraction() {
25         Assertions.assertEquals(5, calc.subtract(10,5));
26     }
27     @Test
28     public void testMultiplication() {
29         Assertions.assertEquals(50, calc.multiply(10,5));
30     }
31     @Test
32     public void testSquare() {
33         Assertions.assertEquals(25, calc.square(5));
34     }
35     @Test
36     public void testCube() {
37         Assertions.assertEquals(27, calc.cube(3));
38     }
39 }
```

Test Results x

Output

```
-----  
Running calculator.CalculatorTest  
Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.095 s -- in calculator.CalculatorTest  
  
Results:  
  
Tests run: 5, Failures: 0, Errors: 0, Skipped: 0  
  
-----  
BUILD SUCCESS  
-----  
Total time: 3.615 s  
Finished at: 2026-04-25T22:08:21+05:00  
-----
```



3. Write Number class having function ispalindrom(), factorial(), number individual digits sum. Write test cases to test the functionality of Number class

Number utils java class


```
10 public class NumberUtils {
11     // Palindrome check
12     public boolean isPalindrome(int num) {
13         int original = num;
14         int reverse = 0;
15         while (num != 0) {
16             int digit = num % 10;
17             reverse = reverse * 10 + digit;
18             num /= 10;
19         }
20         return original == reverse;
21     }
22     // Factorial
23     public int factorial(int num) {
24         int fact = 1;
25
26         for (int i = 1; i <= num; i++) {
27             fact *= i;
28         }
29         return fact;
30     }
31     // Sum of digits
32     public int sumOfDigits(int num) {
33         int sum = 0;
34         while (num != 0) {
35             sum += num % 10;
36             num /= 10;
37         }
38         return sum;
39     }
40 }
```

Transferring Maven repository index: Central R

Number utils test java class

```
Source  History  [Icons]
5      package numbers;
6      /**
7       *
8       * @author user
9       */
10     import numbers.NumberUtils;
11     import org.junit.jupiter.api.Assertions;
12     import org.junit.jupiter.api.BeforeEach;
13     import org.junit.jupiter.api.Test;
14     public class NumberUtilsTest {
15         NumberUtils num;
16         @BeforeEach
17         public void setup() {
18             num = new NumberUtils();
19         }
20         @Test
21         public void testPalindromeTrue() {
22             Assertions.assertTrue(num.isPalindrome(121));
23         }
24         @Test
25         public void testPalindromeFalse() {
26             Assertions.assertFalse(num.isPalindrome(123));
27         }
28         @Test
29         public void testFactorial() {
30             Assertions.assertEquals(120, num.factorial(5));
31         }
32         @Test
33         public void testSumOfDigits() {
34             Assertions.assertEquals(6, num.sumOfDigits(123));
35         }
36     }
37 }
```

Output

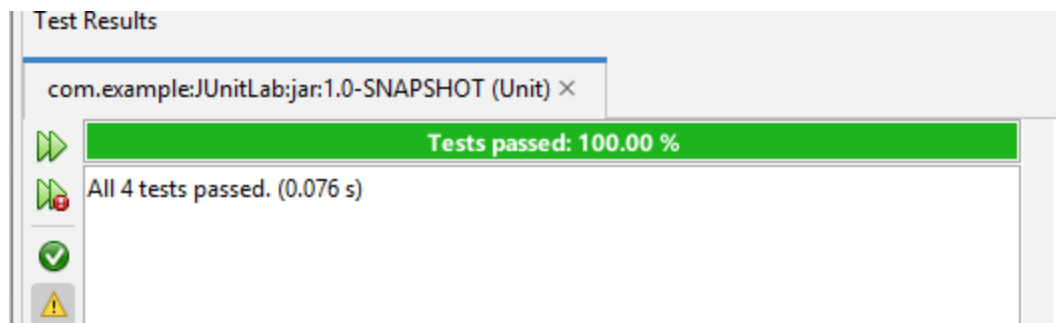
```
-----
Running numbers.NumberUtilsTest
Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.076 s -- in numbers.NumberUtilsTest

Results:

Tests run: 4, Failures: 0, Errors: 0, Skipped: 0

-----
BUILD SUCCESS
-----

Total time: 2.495 s
Finished at: 2026-04-25T22:11:02+05:00
```



All test suite for all classes

```

10  */
11  import org.junit.platform.suite.api.SelectClasses;
12  import org.junit.platform.suite.api.Suite;
13  import triangle.TriangleTest;
14  import calculator.CalculatorTest;
15  import numbers.NumberUtilsTest;
16  @Suite
17  @SelectClasses({
18      TriangleTest.class,
19      CalculatorTest.class,
20      NumberUtilsTest.class
21  })
22  public class AllTestSuite {
23  }

```

Output:

```

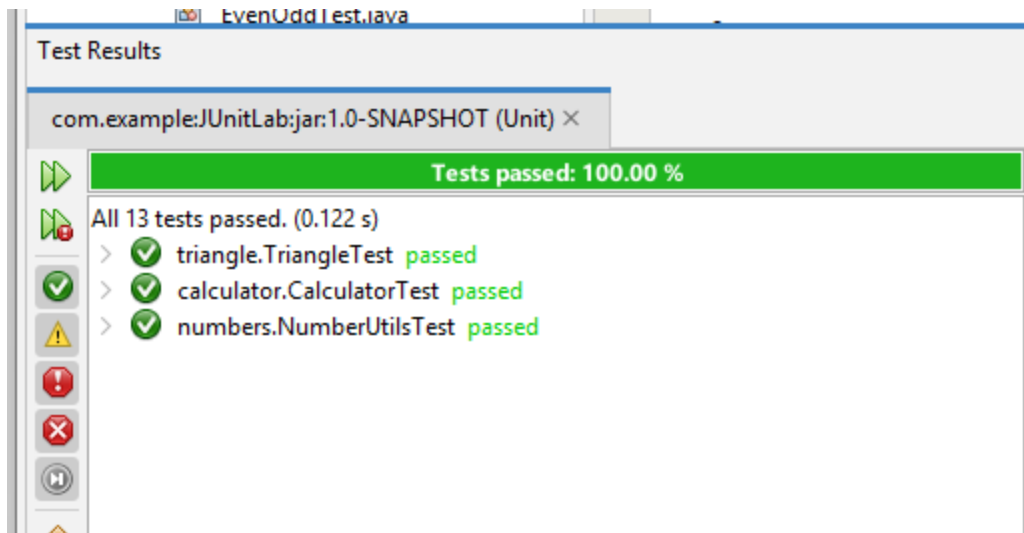
-----
T E S T S
-----
Running AllTestSuite
Running triangle.TriangleTest
Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.075 s -- in triangle.TriangleTest
Running calculator.CalculatorTest
Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.022 s -- in calculator.CalculatorTest
Running numbers.NumberUtilsTest
Tests run: 0, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.175 s -- in AllTestSuite

Results:

Tests run: 13, Failures: 0, Errors: 0, Skipped: 0

-----
BUILD SUCCESS
-----
Total time: 4.127 s
Finished at: 2026-04-25T22:37:34+05:00
-----

```



7.6 Homework: Make a test case for two different classes, one containing a code whether a number is prime or not, second class can contain whether a number is even or odd. Make a test suite of both test classes.

Prime number java class

```
package numbers;

/**
 *
 * @author user
 */
public class PrimeNumber {

    public boolean isPrime(int num) {
        if (num <= 1) return false;

        for (int i = 2; i <= num / 2; i++) {
            if (num % i == 0) {
                return false;
            }
        }
        return true;
    }
}
```

Even odd java class

```

4 | */
5 | package numbers;
6 | /**
7 |  *
8 |  * @author user
9 |  */
10 | public class EvenOdd {
11 |     public boolean isEven(int num) {
12 |         return num % 2 == 0;
13 |     }
14 |     public boolean isOdd(int num) {
15 |         return num % 2 != 0;
16 |     }
17 | }
18 |

```

Prime number test java class

```

5 | package numbers;
6 | /**
7 |  *
8 |  * @author user
9 |  */
10 | import org.junit.jupiter.api.Test;
11 | import static org.junit.jupiter.api.Assertions.*;
12 | public class PrimeNumberTest {
13 |     PrimeNumber p = new PrimeNumber();
14 |     @Test
15 |     void testPrime() {
16 |         assertTrue(p.isPrime(7)); // prime
17 |     }
18 |     @Test
19 |     void testNotPrime() {
20 |         assertFalse(p.isPrime(8)); // not prime
21 |     }
22 | }
23 |

```

Output

```

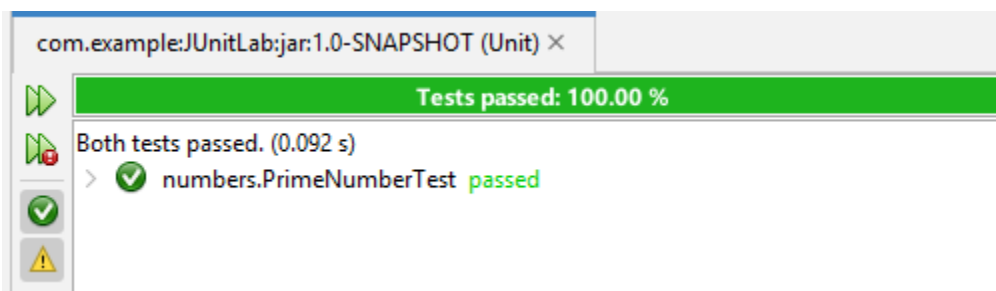
-----
Running numbers.PrimeNumberTest
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.092 s -- in numbers.PrimeNumberTest

Results:

Tests run: 2, Failures: 0, Errors: 0, Skipped: 0

-----
BUILD SUCCESS
-----
Total time: 2.667 s
Finished at: 2026-04-25T22:20:41+05:00
-----

```



Even odd test java class

```

1  /
2
3  package numbers;
4
5  /**
6   *
7   * @author user
8   */
9
10 import org.junit.jupiter.api.Test;
11 import static org.junit.jupiter.api.Assertions.*;
12 public class EvenOddTest {
13     EvenOdd eo = new EvenOdd();
14     @Test
15     void testEven() {
16         assertTrue(eo.isEven(10));
17     }
18     @Test
19     void testOdd() {
20         assertTrue(eo.isOdd(7));
21     }
22 }
23

```

Output

```

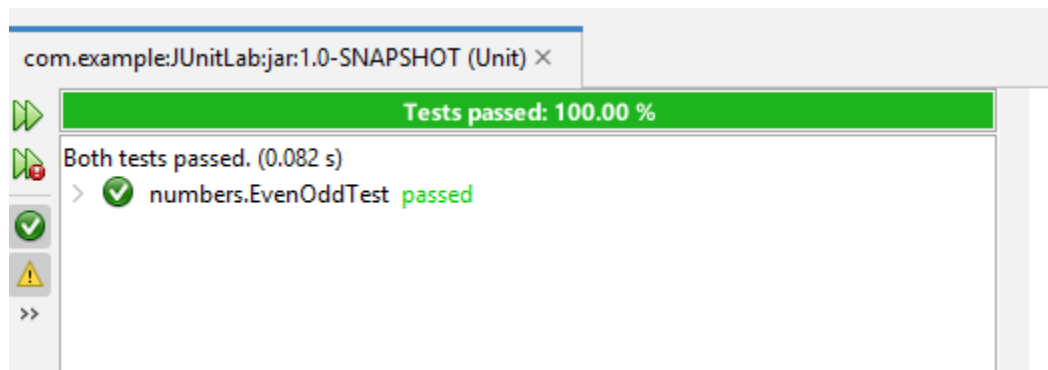
-----
Running numbers.EvenOddTest
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.082 s -- in numbers.EvenOddTest

Results:

Tests run: 2, Failures: 0, Errors: 0, Skipped: 0

-----
BUILD SUCCESS
-----
Total time: 3.873 s
Finished at: 2026-04-25T22:19:54+05:00
-----

```



Pom.xml file code

```

-      </dependency>
-      <!-- Suite API -->
-    </dependencies>
-    <build>
-      <plugins>
-        <plugin>
-          <groupId>org.apache.maven.plugins</groupId>
-          <artifactId>maven-jar-plugin</artifactId>
-          <version>3.1.1</version>
-          <configuration>
-            <archive>
-              <addMavenDescriptor>true</addMavenDescriptor>
-            </archive>
-          </configuration>
-        </plugin>
-      </plugins>
-    </build>
-  </project>

```

All test suit for both classes

```

8  */
9  import org.junit.platform.suite.api.SelectClasses;
0  import org.junit.platform.suite.api.Suite;
1  import numbers.PrimeNumberTest;
2  import numbers.EvenOddTest;
3  @Suite
4  @SelectClasses({
5      PrimeNumberTest.class,
6      EvenOddTest.class
7  })
8  public class AllTestSuite {
9  }
0

```

Output

```

-----
T E S T S
-----
Running AllTestSuite
Running numbers.PrimeNumberTest
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.082 s -- in numbers.PrimeNumberTest
Running numbers.EvenOddTest
Tests run: 0, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.142 s -- in AllTestSuite

Results:

Tests run: 4, Failures: 0, Errors: 0, Skipped: 0

-----
BUILD SUCCESS
-----
Total time: 3.580 s
Finished at: 2026-04-25T22:33:11+05:00
-----

```

com.example:JUnitLab:jar:1.0-SNAPSHOT (Unit) ×








Tests passed: 100.00 %

All 4 tests passed. (0.09 s)

- >  numbers.PrimeNumberTest passed
- >  numbers.EvenOddTest passed